

JAMES CHIBUIKE OKORIE

Embedded Systems Laboratory, Department of Physics, University of Cape Coast, Cape Coast, CR, Ghana

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Professional Profile

An embedded systems and robotics engineer in training with an MPhil in Physics (Embedded Systems and Automation) in progress, experienced in Arduino/ESP32-based prototyping, IoT systems, and sensor integration for smart agriculture and automation projects. I have contributed to solutions such as a solar-powered thermal-imaging system for Fall Armyworm detection, Arduino-based fruit-defect detection, soil-parameter monitoring, low-cost hatching systems, and automated titration setups, taking projects from concept through hardware design, firmware development, and data acquisition to user-oriented outputs. I am particularly motivated by using physics, IoT, and robotics to build scalable, low-cost tools for learning, precision farming, and innovation in African contexts.

Education

Master of Philosophy in Physics (Embedded Systems and Automation) Expected Mar 2026

University of Cape Coast

Cape Coast, CR

- **Related Coursework:** Advanced Quantum Mechanics, Computing and Numerical Methods, Quantum Statistics, Mathematical Methods, Microprocessor I and II, Graduate Seminar, Academic Writing
- Research Assistant to Prof. Kwadwo Anokye Dompheh at the Embedded System Laboratory

Bachelor of Science in Engineering Physics

Awarded Nov 2022

University of Cape Coast

Cape Coast, CR

- **Related Coursework:** Data Acquisition Systems, Quantum Mechanics I&II, Digital Electronics, Classical Mechanics, Computing for Physics I&II, Further Statistics, Advanced Calculus, Material Science I&II
- **Undergraduate Thesis:** "RFID-BASED EXAMINATION ATTENDANCE SYSTEM"

SKILLS

Technical Skills & Tools

- **Programming:** Python, C++, RStudio, MicroPython, MATLAB, Simulink, Arduino IDE.
- **Machine Learning:** TensorFlow, Keras, Scikit-learn, TinyML, CNNs, reinforcement learning.
- **Robotics & Hardware:** Arduino (ESP32, Nano RP2040, Nano BLE Sense 33, Uno, Mega 2560), Micro:bit, STM32, sensor interfacing, LoRa, WAN, MQTT, ROS 1 & 2 (self-learned), Circuit Design, PCB Prototyping.
- **Software:** LabVIEW, Fritzing, Proteus, FreeCAD, Overleaf LaTeX, Git/GitHub.

Research & Analytical Skills

- Scientific Research Methodologies, Experimental Design & Implementation, Data Acquisition & Interpretation, Problem-Solving & Critical Thinking, Literature Review & Synthesis, Technical Report Writing, Statistical Analysis.

Soft Skills & Professional Attributes

- Teamwork & Collaboration, Leadership & Mentorship, Effective Communication (Written & Oral), Time Management & Organization, Adaptability & Fast Learning, Project Management, Presentation Skills.

Research Experience

Research Scholar, Intra-African PATH Project

Feb 2025 – Present

Genetics, Biotechnology & Seed Sciences Laboratory (GBioS),

University of Abomey-Calavi

Abomey-Calavi, Atlantique

- Developing a solar-powered thermal imaging system for the early detection of Fall Armyworm in maize as part of a research team.
- Engaged in advanced research on intelligent agricultural systems, contributing to climate-smart solutions.

Research Assistant

Nov 2022 – Present

Embedded Systems Laboratory, Department of Physics, University of Cape Coast

Cape Coast, CR

Topics: Arduino-Based Orange Fruit Defect Detection Using Resistance And Two-Point Probe Measurement

- Prototyped and integrated an innovative Arduino-based orange fruit defect detection system, gaining proficiency in embedded systems, data analysis, and statistical modeling.

Other Research Project Assisted

Nov 2022 – Present

Embedded Systems Laboratory, Department of Physics, University of Cape Coast

Cape Coast, CR

- Estimation Of Nitrogen N, Phosphorus P And Potassium in the Soil Using Regression Analysis
- Monitoring of Soil moisture content using a resistive moisture sensor via Arduino
- A study of the design and construction of a Low-cost egg Hatching System

- Simulation and Testing of An Arduino-Based Titration System

Professional Experience

Data Science and Machine Learning Intern

April 2025 – Present

Edureka

Online

- Gaining hands-on experience in data collection, preprocessing, and machine learning model development, utilizing Python and key ML libraries.
- Actively learning and applying data analysis, statistical modeling, and visualization techniques under guidance.

Laboratory Assistant

Nov 2022 – Present

Embedded Systems Laboratory, Department of Physics, University of Cape Coast

Cape Coast, CR

- Organized practical project sessions and assisted students with their projects and practical coursework.
- Supported lecturers in practical teaching of undergraduates, planning departmental research, ensuring lab procedures, and participating in project discussions.

Robotics Engineer Lead

Oct 2023 – Present

Tech4Everyone

Cape Coast, CR

- Designed and delivered robotics courses, created YouTube tutorials, and guided students on STEM projects.

Co-Founder & Technical Lead

2023 – Present

Low Cost tPBM Cape Design, Sunshine PBM Wellness Center

- Lead the design and development of an innovative Low-Cost Transcranial Photo Biomodulation (tPBM) cap for brain injury and wellness

Tutor, Programming with Python

July 2024 – Feb 2025

Latter-Day Saints, Abura Stake YSA

Cape Coast, CR

- Delivered comprehensive lessons on Python programming fundamentals (syntax, control flow, data structures, project development) to foster practical coding skills.
- Mentored students through hands-on exercises and collaborative projects, enhancing problem-solving abilities and practical application of coding concepts.

Demonstrator

Oct 2024 – Feb 2025

Department of Physics, University of Cape Coast

Cape Coast, CR

- Assisted undergraduate students during practical laboratory sessions and provided academic support for coursework.
- Represented my supervisor in lectures and facilitated learning in their absence.
- Contributed to the assessment process through invigilation, grading, and marking of quizzes and scripts.

Teaching Assistant

Nov 2022 – Jan 2025

Department of Physics, University of Cape Coast

Cape Coast, CR

- Assisted lecturers in teaching undergraduate students, organizing research data, and planning departmental research.
- Designed assessment tools and provided tutoring/mentorship to enhance student learning and progress.

Certifications and Professional Development

Data Science for Research and Institutional Intelligence | RUFORUM

Aug 11-15, 2025

- Explored predictive, prescriptive, and descriptive analytics.
- Gained experience with integrating machine learning with QGIS and model deployment with R Shiny

Data Management for Research and Institutional Decision-Making | RUFORUM

Jul 21-25, 2025

- Trained on practical tools, including RStudio for data management and exploratory analysis, QGIS for spatial data visualization, and Git/GitHub for collaborative data sharing and storage.

Scientific writing for Early Career Scientists | RUFORUM

Jul 14-18, 2025

- Focused on scientific article structure, avoiding plagiarism, and identifying predatory journals, enhancing skills for manuscript preparation.

eLwazi Open Data Science Platform (ODSP) Research Data Management Course

Jun 23-Jul 16, 2025

- Completed training on effective research data management principles for data scientists in Africa.

Unpublished Research Work

- **Okorie, J.C. (2024).** Simulation and Testing of An Arduino-Based Titration System. Unpublished Manuscript, Embedded Systems Laboratory, Department of Physics, University of Cape Coast, Cape Coast, CR, Ghana.
- Frimpong, E., **Okorie, J.C.**, & Dompheh, K.A. **(2023).** Arduino-Based Orange Fruit Defect Detection Using Resistance and Two-Point Probe Measurement. Unpublished Manuscript, Embedded Systems Laboratory, Department of Physics, University of Cape Coast, Cape Coast, CR, Ghana.
- **Okorie, J.C. (2022).** RFID-Based Examination Attendance System. Unpublished Manuscript, Department of Physics, University of Cape Coast, Ghana.

Selected Conferences & Workshops

- 3rd African Conference on Precision Agriculture (AfCPA), Precision Agriculture in Action for Africa, University of Cape Coast, Ghana, Satellite Station **Satellite Panellist & Participant**
- Ghana-Norway Summer School in Medical Physics and Radiography, Emergence of New Technologies in Diagnostic Imaging, University of Cape Coast. **16th - 20th October 2023, Participant**
- Low-cost Optical Spectroscopy Techniques for Vector (Mosquito) Surveillance, Laser and Fiber Optic Center (LAFOC), University of Cape Coast. **21st - 25th August 2023, Participant**

Positions & Responsibilities

Community Manager **Feb 2022 - Present**
ARM (E³) NGAGE Student Club, University of Cape Coast (UCC ARM-y)

Advisor **Jan 2025 - Present**
National Society of Black Engineers (NSBE-UCC)

President **Nov 2022 - Present**
International Movement of Catholic Students Union (IMCS-PAX ROMANA Alumni, UCC Local)

General Secretary **Oct 2021 – Mar 2023**
National Society of Black Engineers (NSBE-UCC)

President **Oct 2021 – Oct 2022**
Knights and Ladies of the Altar (KNOLTA), OLSWCC-UCC

Professional Affiliations

- African Association for Precision Agriculture (AAPA) **May 2025-Present**
- UCC ARM-y, arm (E³) NGAGE Student Club, University of Cape Coast, Cape Coast, Ghana **Feb 2022-Present**
- Physics Students Association of Ghana (PHYSAG-UCC), University of Cape Coast, Ghana **Aug 2018 - Present**
- National Society of Black Engineers (NSBE-UCC), University of Cape Coast, Ghana **Aug 2018 – Nov 2022**
- International Movement of Catholic Students Union (IMCS-PAX ROMANA, UCC Local) **Aug 2018 - Nov 2022**

Awards And Honors

Executive Member of the Year **Awarded Dec 2022**
Dinner and Awards Night 2024, International Movement of Catholic Student Union (IMCS-PAX, ROMANA, UCC)

REFEREES

Available upon request.